

Marcus Acosta

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EXPERIENCE

Bettorca

Remote

Co-Founder & Engineer

Sep 2025 - Present

- Developed **XGBoost** and **PyTorch GRU** models for outcome predictions, reducing Brier error by **10.1%** across **14K+** held-out evaluations.
- Conducted feature-set and ensemble weighting experiments, measuring a **+0.009 ROC-AUC** improvement within a **95%** paired-bootstrap interval.
- Engineered **50K+** historical game records into time-aware inputs, adding automated tests for leakage, temporal ordering, and data integrity.
- Architected production inference pipelines integrating **FastAPI** into **Go**, delivering **100+** daily predictions across **20+** live production environments.
- Implemented reproducibility protocols using **SHA-256** artifact hashing, replicating **900+** metrics with fail-closed checks.
- Integrated **MLflow** monitoring to track model versions, performance metrics, artifacts, and prediction latency across production inference.

RESEARCH PROJECTS

LitGPT

GitHub

Multi-Turn Supervised Fine-Tuning & Loss Masking

- Improved multi-turn LLM fine-tuning by implementing assistant-focused training across complete conversations while preserving existing training behavior.
- Evaluated conversational fine-tuning through controlled experiments measuring response quality, context retention, and general capability tradeoffs.

torchTitan

GitHub

Mixture-of-Depths Auxiliary Loss Normalization

- Improved DeepSeek model training by implementing per-depth loss normalization across distributed training while preserving existing optimization behavior.
- Analyzed optimization effects across gradient scaling, padding, and variable-length sequences to identify when normalization improves training consistency.

SKILLS

Languages: Python, CUDA, Go, Rust, Bash, SQL

Frameworks & Libraries: PyTorch, XGBoost, Hugging Face (Transformers, Tokenizers), FastAPI

Distributed Training: PyTorch Distributed (Distributed, DeviceMesh), Lightning Fabric

Infrastructure: Docker, Kubernetes, ONNX, MLflow, GitHub Actions, Railway

Data: PostgreSQL, Qdrant, Redis, pandas, PyArrow

ADDITIONAL EXPERIENCE

LifeTime Inc.

Walnut Creek, CA

Facility Operations Supervisor

Jan 2024 - June 2026

- Led daily operations for a 5–7 person facilities team, coordinating schedules, vendors, inventory, safety inspections, and compliance documentation.

EDUCATION

Post-Baccalaureate Coursework in Computer Science

Oregon State University

- Data Structures, Discrete Mathematics, Databases, Software Engineering, Web Development

Bachelor of Arts, Sociology

UC Santa Cruz